EXPERIMENT NO. 8 (Group B)

Aim: Data Cleaning and Preparation Problem Statement: Analyzing Customer Churn in a Telecommunications Company Dataset: "Telecom_Customer_Churn.csv" Description: The dataset contains information about customers of a telecommunications company and whether they have churned (i.e., discontinued their services). The dataset includes various attributes of the customers, such as their demographics, usage patterns, and account information. The goal is to perform data cleaning and preparation to gain insights into the factors that contribute to customer churn.

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Outputs:

Step 1 — Import and Load Dataset

In [13]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv("Telco_Customer_Churn_Expanded.csv")

print("Dataset loaded successfully!")
print("Shape:", data.shape)
display(data.head())
```

```
Dataset loaded successfully!
Shape: (15043, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleL
0	7590-VHVEG	Female	0	Yes	No	1	No	No ph se
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No ph se
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns

Step 2 — Explore the Dataset

In [14]:
```
print("Dataset Information:")
data.info()

print("\nColumn Names:")
print(data.columns.tolist())

print("\nStatistical Summary:")
display(data.describe())

print("\nUnique Values:")
print(data.nunique())
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15043 entries, 0 to 15042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        15043 non-null  object
 1   gender            15043 non-null  object
 2   SeniorCitizen     15043 non-null  int64
 3   Partner           15043 non-null  object
 4   Dependents        15043 non-null  object
 5   tenure            15043 non-null  int64
 6   PhoneService      15043 non-null  object
 7   MultipleLines     15043 non-null  object
 8   InternetService   15043 non-null  object
 9   OnlineSecurity    15043 non-null  object
 10  OnlineBackup      15043 non-null  object
 11  DeviceProtection  15043 non-null  object
 12  TechSupport       15043 non-null  object
 13  StreamingTV       15043 non-null  object
 14  StreamingMovies   15043 non-null  object
 15  Contract          15043 non-null  object
 16  PaperlessBilling  15043 non-null  object
 17  PaymentMethod     15043 non-null  object
 18  MonthlyCharges    15043 non-null  float64
 19  TotalCharges      15043 non-null  object
 20  Churn             15043 non-null  object
dtypes: float64(1), int64(2), object(18)
memory usage: 2.4+ MB

Column Names:
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneS
ervice', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'Devi
ceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'Paperle
ssBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']

Statistical Summary:
```

	SeniorCitizen	tenure	MonthlyCharges
count	15043.000000	15043.000000	15043.000000
mean	0.162733	32.717144	66.782433
std	0.369135	24.586526	29.711526
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	41.602308
50%	0.000000	29.000000	69.780892
75%	0.000000	56.000000	91.610379
max	1.000000	72.000000	118.750000

```
Unique Values:
customerID          7043
gender                 2
SeniorCitizen          2
Partner                2
Dependents             2
tenure                73
PhoneService           2
MultipleLines          3
InternetService        3
OnlineSecurity         3
OnlineBackup           3
DeviceProtection       3
TechSupport            3
StreamingTV            3
StreamingMovies        3
Contract               3
PaperlessBilling       2
PaymentMethod          4
MonthlyCharges      9585
TotalCharges        6531
Churn                  2
dtype: int64
```

Step 3 — Handle Missing Values, Duplicates & Data Types

In [15]:
```
# Convert TotalCharges into numeric format
data["TotalCharges"] = pd.to_numeric(
    data["TotalCharges"], errors="coerce"
)

print("Missing values before handling:")
print(data.isnull().sum())

# Fill missing TotalCharges with median
data["TotalCharges"] = data["TotalCharges"].fillna(
    data["TotalCharges"].median()
)
```

```
print("\nMissing values after handling:")
print(data.isnull().sum())
```

```
Missing values before handling:
customerID           0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges        19
Churn                0
dtype: int64

Missing values after handling:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

Step 4 — Outlier Detection & Visualization

In [16]:
```
print("Duplicate rows before removal:", data.duplicated().sum())

data.drop_duplicates(inplace=True)
```

```
print("Duplicate rows after removal:", data.duplicated().sum())

# Remove unwanted spaces from column values
for col in data.select_dtypes(include="object").columns:
    data[col] = data[col].str.strip()

print("\nDataset shape after cleaning:", data.shape)
```

```
Duplicate rows before removal: 0
Duplicate rows after removal: 0

Dataset shape after cleaning: (15043, 21)
```

Step 5 — Train-Test Split & Export Cleaned Dataset

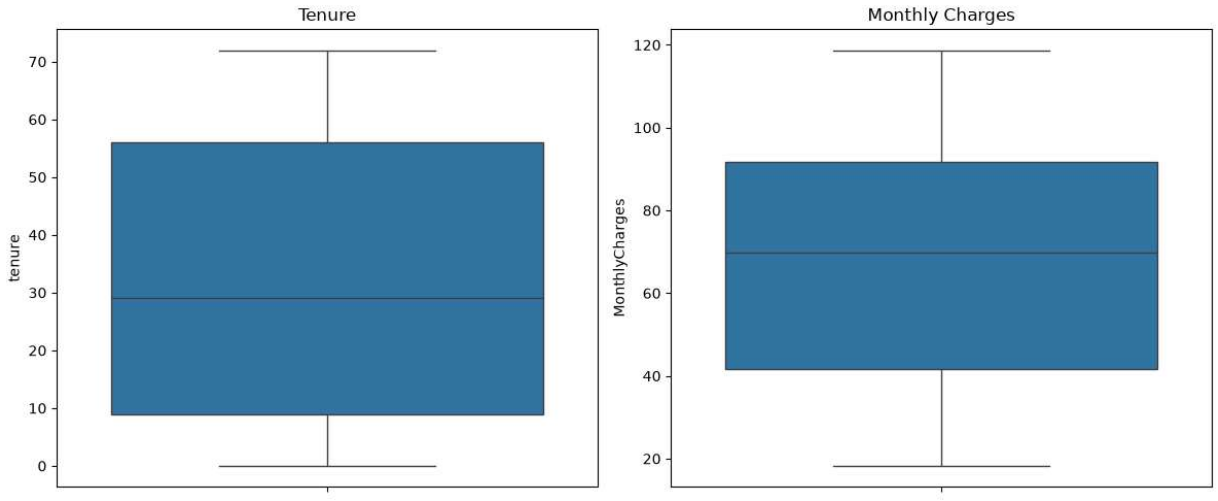
In [18]:
```
# Boxplots for numerical columns

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
sns.boxplot(y=data["tenure"])
plt.title("Tenure")

plt.subplot(1, 2, 2)
sns.boxplot(y=data["MonthlyCharges"])
plt.title("Monthly Charges")

plt.tight_layout()
plt.show()
print("Name:Aditya Rade Roll No:BEAD23167")
```




```
Name:Aditya Rade Roll No:BEAD23167
```